

Relationship Between the Sensory Processing Patterns and the Detection Threshold of Curvature Gain

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ABSTRACT

This study examines the relationship between sensory processing patterns and the effects of redirected walking (RDW). Research efforts have been devoted to identifying the detection threshold (DT) of the RDW techniques, and various DTs have been reported in different studies. Recently, age, sex, and spatial ability have been found to be associated with the DTs of RDW techniques. A preliminary examination was conducted on the relationship between sensory processing patterns, as measured by the Adolescents/Adult Sensory Profile, and the DT of curvature gains, one of the fundamental RDW techniques, and it was suggested that the higher sensory sensitivity tendencies were associated with lower DT, i.e., participants were more likely to notice the RDW technique.

Index Terms: Human-centered computing—Human-computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Redirected walking (RDW) is a method that allows a potentially vast virtual space to be compressed into a narrow real space. It has been shown that users experience a detection threshold (DT) for RDW, such that, when RDW manipulation exceeds the DT, the user notices the manipulation, leading to a decrease in presence and cybersickness. In addition to gender and age, spatial cognitive ability has been reported to be related to the DT of RDW technology [3,4]. Moreover, the spatial ability is affected by the neurological differences. For example, individuals with autism spectrum disorders were found to exhibit superior performance compared to a control group for certain spatial abilities [5]. Thus, neurological characteristics might be closely related to the effects of RDW.

Dunn's model of sensory processing was proposed as a theory of neurological characteristics [2]. It provides a broad interpretation of the individual behavior by considering neurological thresholds and behavioral responses/self-regulation as a mutually interactive continuum. This study examines the relationship between the DT of RDW and sensory processing patterns using the Adolescent/Adult Sensory Profile (AASP), which was designed based on Dunn's model of sensory processing [1]. The AASP is the most widely applied sensory assessment for adolescent and adults, and generates a personal profile for sensory processing across four quadrants—low registration, sensation seeking, sensory sensitivity, and sensation avoidance.

2 USER STUDY

This study investigated whether an association existed between the RDW thresholds and the neurological characteristics. The study employed a pseudo-two-alternative forced choice (2AFC) task to evaluate the DTs of curvature gain and the AASP to interpret neurological characteristics.

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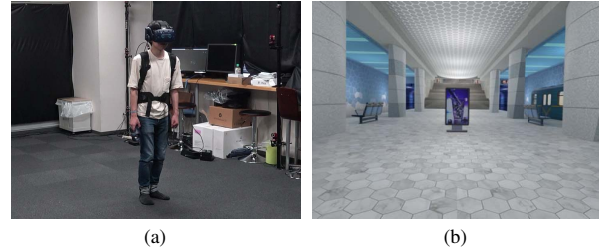


Figure 1: Experimental setup: (a) Participant wears an HMD and a Backpack PC; (b) the virtual environment during the experiment.

Table 1: Number of participants classified into five-class classification for the four quadrants of the AASP. No participants were classified as "much less," so "much less" column was omitted.

	Much more	More	Similar	Less
Low Registration	1	1	16	-
Sensation Seeking	1	4	12	1
Sensory Sensitivity	1	5	12	-
Sensation Avoiding	2	4	11	1

2.1 Participants

Due to the spread of COVID-19, the participants in the experiment comprised local campus students and staff with ages ranging from 20 to 65 years old. Among them, 18 participants completed the experiment, and were included in the analyses (3 females and 15 males; age: $M = 25.3$, $SD = 4.01$; body height: $M = 1.69$ m, $SD = 0.066$ m). The participants received 1,500 Japanese Yen (approximately 13 USD) Amazon gift cards in return for their participation.

2.2 Experimental environment

The experiment was conducted in a 6 m $\times$ 8 m laboratory room. The participants wore a HP VR Backpack G2 that contained an Intel Core i7 vPro processor, 32 GB of main memory, and an NVIDIA GeForce RTX 2080 graphics card. They also wore an HTC Vive Pro Eye head-mounted display (HMD) and held Vive Pro controllers (see Fig.1(a)). The virtual environment was rendered by using the Unity 2019.4, and included a space that reproduced a subway station in front of the participant. A poster stand was placed to indicate the target toward which the user should walk (see Fig.1(b)).

2.3 Procedure

Before the experiment, all participants completed an informed consent form and answered the AASP. They then received a detailed explanation of the experiment and instructions on how to perform the experimental tasks. The participants completed training trials prior to the actual experimental trials. Nine curvature gains were used in this experiment included $g_C \in \{-0.2, -0.1, -0.05, -0.025, 0, 0.025, 0.05, 0.1, 0.2\}$. The participants were randomly presented with the above nine conditions. Each

Table 2: The pooled results of the point of subjective equality (PSE) and detection threshold (DT) for the four quadrants of the AASP. The values in parentheses serve as a reference.

Low Registration				Sensation Seeking			
	Much more	More	Similar	Much more	More	Similar	Less
PSE	(-.005)	(-.013)	-.002	(-.015)	-.020	.005	(-.011)
DT	(.023)	(.072)	.043	(.047)	.046	.045	(.024)

Sensory Sensitivity				Sensation Avoiding			
	Much more	More	Similar	Much more	More	Similar	Less
PSE	(-.005)	-.007	.000	.015	-.021	.000	(.003)
DT	(.023)	.037	.048	.033	.042	.048	(.020)

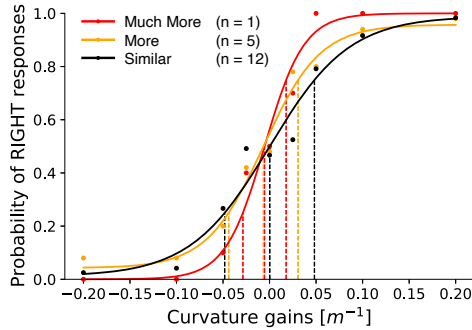


Figure 2: Psychometric functions fitted according to the sensory sensitivity on the AASP. The horizontal axis represents the value of curvature gain, and the vertical axis represents the probability of answering “I felt walking to the right.”

condition was presented 10 times; thus, each participant performed a total of 90 trials. In each trial, the participant walked toward the target (the poster stand) for 6.5 m, while the curvature gain was implemented for the last 5 m. Subsequently, the participants were asked to answer the following questionnaire: “Did you feel like walking to the left or to the right?”, which was designed for a pseudo-2AFC task, and answered it using the controllers.

3 RESULTS

3.1 AASP Scores

The raw scores for each quadrant was calculated, and the data samples were classified into an appropriate five-class classification, including categories of “much more than,” “more than,” “similar to,” “less than,” and “much less than” most people based on the cut scores. Table 1 shows the number of participants classified into five-class classification for the four quadrants of the AASP. In this experiment, no participants were classified as “much less,” so “much less” column was omitted.

3.2 DTs of Curvature Gain

In the experiment, three participants provided responses that were completely opposite to their recorded behaviors, i.e., answering that they walked to the left when they walked to the right. These participants’ responses were inverted and analyzed. Fig. 2 shows the pooled results of the 2AFC task for sensory sensitivity on the AASP. The x-axis shows the applied curvature gains. The y-axis indicates the probability of answering “I felt walking to the right”. Each curve was fitted with a sigmoidal psychometric function to determine the DT. Table 2 lists the point of subjective equalities (PSEs) and DTs for the four quadrants of the AASP. The values in parentheses can be used as a reference because only one person was sampled.

4 DISCUSSION

This experiment aimed to identify an association between the RDW thresholds and the sensory processing patterns. The pooled results for each quadrant of the AASP showed differences in sensory sensitivity and sensation avoiding between the columns. For sensory sensitivity, the DT of the participants who were similar to most people ($n=12$) was $g_C = .048$ and those participants who were more sensitive than most people ($n=5$) was $g_C = .037$. The results suggest that the threshold was 22% lower for those who were prone to sensory sensitivity in comparison to most people. For sensation avoiding, the DT of the participants who were similar to most people ($n=11$) was $g_C = .048$, that for those who were more sensation-avoidance than most people ($n=4$) was $g_C = .042$, and that for those who were much more so than most people ($n=2$) was $g_C = .033$. The results indicate that the threshold was 31% lower for those who were extremely prone to sensation avoidance and 13% lower for those who were prone to sensation avoidance compared to most people. Thus, the DT of curvature gain tended to be lower in the group with higher sensory sensitivity and sensation avoidance tendencies, i.e., lower neurological threshold.

5 CONCLUSION

This study evaluated the relationship between the DT of curvature gain and the sensory processing patterns. AASP was used to evaluate the neurological thresholds, and a psychophysical experiment was conducted to estimate the participants’ DTs of curvature gain. Our results suggested that the participants with a higher propensity for sensory sensitivity and sensation avoidance were more likely to notice the RDW techniques.

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